Assignment 3: Naive Bayes Classification

Robert Andoh

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Introduction

This assignment demonstrates the application of the Naive Bayes classification algorithm using the Universal Bank dataset.

The objective is to estimate the probability that a customer accepts a personal loan offer based on two predictor variables:

- Online Banking Usage (Online)
- Credit Card Ownership (CreditCard)

The analysis includes data partitioning, probability calculations, pivot tables, and Naive Bayes model implementation.

Load Required Packages

```
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
library(e1071)
```

```
##
## Attaching package: 'e1071'
```

```
## The following object is masked from 'package:ggplot2':
##
##     element
```

Import Dataset

```r
bank <- read.csv("UniversalBank-1.csv")

head(bank)
```

```
##   ID Age Experience Income ZIP.Code Family CCAvg Education Mortgage
## 1  1  25          1     49    91107      4   1.6         1        0
## 2  2  45         19     34    90089      3   1.5         1        0
## 3  3  39         15     11    94720      1   1.0         1        0
## 4  4  35          9    100    94112      1   2.7         2        0
## 5  5  35          8     45    91330      4   1.0         2        0
## 6  6  37         13     29    92121      4   0.4         2      155
##   Personal.Loan Securities.Account CD.Account Online CreditCard
## 1             0                  1          0      0          0
## 2             0                  1          0      0          0
## 3             0                  0          0      0          0
## 4             0                  0          0      0          0
## 5             0                  0          0      0          1
## 6             0                  0          0      1          0
```

```r
dim(bank)
```

```
## [1] 5000   14
```

```r
str(bank)
```

```
## 'data.frame':    5000 obs. of  14 variables:
##  $ ID                : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Age               : int  25 45 39 35 35 37 53 50 35 34 ...
##  $ Experience        : int  1 19 15 9 8 13 27 24 10 9 ...
##  $ Income            : int  49 34 11 100 45 29 72 22 81 180 ...
##  $ ZIP.Code          : int  91107 90089 94720 94112 91330 92121 91711 93943 90089 93023 ...
##  $ Family            : int  4 3 1 1 4 4 2 1 3 1 ...
##  $ CCAvg             : num  1.6 1.5 1 2.7 1 0.4 1.5 0.3 0.6 8.9 ...
##  $ Education         : int  1 1 1 2 2 2 2 3 2 3 ...
##  $ Mortgage          : int  0 0 0 0 0 155 0 0 104 0 ...
##  $ Personal.Loan     : int  0 0 0 0 0 0 0 0 0 1 ...
##  $ Securities.Account: int  1 1 0 0 0 0 0 0 0 0 ...
##  $ CD.Account        : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ Online            : int  0 0 0 0 0 1 1 0 1 0 ...
##  $ CreditCard        : int  0 0 0 0 1 0 0 1 0 0 ...
```

Prepare Variables for Naive Bayes

```r
bank_nb <- bank[, c("Online", "CreditCard", "Personal.Loan")]

names(bank_nb) <- c("Online", "CC", "Loan")

bank_nb$Online <- factor(bank_nb$Online)
```

```r
bank_nb$CC <- factor(bank_nb$CC)
bank_nb$Loan <- factor(bank_nb$Loan)

head(bank_nb)
```

```
##   Online CC Loan
## 1      0  0    0
## 2      0  0    0
## 3      0  0    0
## 4      0  0    0
## 5      0  1    0
## 6      1  0    0
```

Data Partitioning

```r
set.seed(123)

train_index <- sample(
  1:nrow(bank_nb),
  size = 0.60 * nrow(bank_nb)
)

train <- bank_nb[train_index, ]

validation <- bank_nb[-train_index, ]

dim(train)
```

```
## [1] 3000    3
```

```r
dim(validation)
```

```
## [1] 2000    3
```

Question A: Pivot Table for Training Data

```r
pivot_A <- xtabs(
  ~ CC + Loan + Online,
  data = train
)

pivot_A
```

```
## , , Online = 0
##
##    Loan
```

```
## CC     0    1
##   0  785   65
##   1  317   34
##
## , , Online = 1
##
##    Loan
## CC     0    1
##   0 1145  122
##   1  475   57
```

```
dim(bank_nb)
dim(train)
dim(validation)

set.seed(123)

train_index <- sample(
  1:nrow(bank_nb),
  size = 0.60 * nrow(bank_nb)
)

train <- bank_nb[train_index, ]

validation <- bank_nb[-train_index, ]

dim(train)
dim(validation)

``` r
pivot_A <- xtabs(
  ~ CC + Loan + Online,
  data = train
)

pivot_A
```

```
## , , Online = 0
##
##    Loan
## CC     0    1
##   0  785   65
##   1  317   34
##
## , , Online = 1
##
##    Loan
## CC     0    1
##   0 1145  122
##   1  475   57
```
```

Question B: Posterior Probabilities

```
prop.table(pivot_A)
```

```
## , , Online = 0
##
##    Loan
## CC           0          1
##   0 0.26166667 0.02166667
##   1 0.10566667 0.01133333
##
## , , Online = 1
##
##    Loan
## CC           0          1
##   0 0.38166667 0.04066667
##   1 0.15833333 0.01900000
```

Question C: Separate Pivot Tables

```
pivot_online <- xtabs(
  ~ Loan + Online,
  data = train
)

pivot_online
```

```
##     Online
## Loan    0    1
##    0 1102 1620
##    1   99  179
```

```
pivot_cc <- xtabs(
  ~ Loan + CC,
  data = train
)

pivot_cc
```

```
##     CC
## Loan    0    1
##    0 1930  792
##    1  187   91
```

Question D: Conditional and Prior Probabilities

```r
P_CC1_given_Loan1 <- 91 / (187 + 91)

P_Online1_given_Loan1 <- 179 / (99 + 179)

P_Loan1 <- 278 / 3000

P_CC1_given_Loan0 <- 792 / (1930 + 792)

P_Online1_given_Loan0 <- 1620 / (1102 + 1620)

P_Loan0 <- 2722 / 3000

prob_table <- data.frame(
  Probability = c(
    "P(CC=1|Loan=1)",
    "P(Online=1|Loan=1)",
    "P(Loan=1)",
    "P(CC=1|Loan=0)",
    "P(Online=1|Loan=0)",
    "P(Loan=0)"
  ),
  Value = c(
    P_CC1_given_Loan1,
    P_Online1_given_Loan1,
    P_Loan1,
    P_CC1_given_Loan0,
    P_Online1_given_Loan0,
    P_Loan0
  )
)

prob_table
```

```
##          Probability      Value
## 1     P(CC=1|Loan=1) 0.32733813
## 2 P(Online=1|Loan=1) 0.64388489
## 3          P(Loan=1) 0.09266667
## 4     P(CC=1|Loan=0) 0.29096253
## 5 P(Online=1|Loan=0) 0.59515062
## 6          P(Loan=0) 0.90733333
```

Question E: Manual Naive Bayes Calculation

```r
numerator_loan1 <-
  P_CC1_given_Loan1 *
  P_Online1_given_Loan1 *
  P_Loan1

numerator_loan0 <-
  P_CC1_given_Loan0 *
```

```
  P_Online1_given_Loan0 *
  P_Loan0

manual_nb_probability <-
  numerator_loan1 /
  (numerator_loan1 + numerator_loan0)

manual_nb_probability
```

```
## [1] 0.1105637
```

Question F: Compare Pivot Table and Naive Bayes Results

```
pivot_probability <- 57 / (475 + 57)

comparison_table <- data.frame(
  Method = c(
    "Pivot Table Estimate",
    "Naive Bayes Estimate"
  ),
  Probability = c(
    pivot_probability,
    manual_nb_probability
  )
)

comparison_table
```

```
##                  Method Probability
## 1 Pivot Table Estimate   0.1071429
## 2 Naive Bayes Estimate   0.1105637
```

‘‘‘

Interpretation

The pivot table estimate is obtained directly from the observed frequencies of customers with Online = 1 and CC = 1.

The Naive Bayes estimate is calculated using conditional probabilities and the assumption that Online Banking usage and Credit Card ownership are conditionally independent given loan acceptance.

The pivot table estimate is:

57 / (475 + 57) = 0.1071 (10.71%)

The Naive Bayes estimate is:

0.1106 (11.06%)

The two estimates are very similar, indicating that the Naive Bayes independence assumption works reasonably well for this dataset.

Discussion and Conclusion

This assignment applied the Naive Bayes classification method to predict personal loan acceptance using Online Banking usage and Credit Card ownership.

The dataset was partitioned into training and validation datasets using a 60:40 split. Conditional and prior probabilities were calculated from the training data and used to estimate the probability that a customer would accept a personal loan.

The pivot table approach estimated the probability at approximately 10.71%, while the Naive Bayes classifier estimated the probability at approximately 11.06%.

The similarity between the two estimates suggests that the Naive Bayes conditional independence assumption is reasonable for this dataset.

These findings demonstrate the usefulness of probability-based machine learning methods in customer targeting, lending decision support, and predictive analytics.